PROJECT-BASED LEARNING

PHASE-II EVALUATION

Detective Mystery: The Hidden Truth

Team ID: DSCPP-III-2026-T152

Department of Computer Science & Engineering
Graphic Era (Deemed to be University), Dehradun
Academic Session 2026–27

Team & Mentor Details

Team Information

Team ID: T152
Semester: 3rd
Section: DS2,F,J,K
Domain: *Game Development*

Team Members

1. *Karan Badhani – 2510360037*
2. *Khushi Singal – 2510360039*
3. *Suraj Giri Goswami – 2510012585*
4. *Tanmay Arora – 2510610178*

Mentor

Prof. Dr. Jyoti Agarwal

Interactions completed: 4

Karan leads the team and the game flow, Khushi built the map, Suraj with the player and history, Tanmay with the clues and suspects.

Phase-I Recap & Feedback Incorporated

Phase-I Status

In Phase-I we proposed a C++ detective game that uses OOP and Data Structures. We completed the proposal, the design and the plan but we had no working code yet.

Feedback Received

- *Too many data structures listed with no clear use*
- *Individual contributions were not specific*
- *Progress was only planning with no implementation*

Changes Incorporated

- *Four data structures each with one job*
- *One module per member with own commits*
- *A working model with the story set on our campus*

Problem Statement & Finalized Objectives

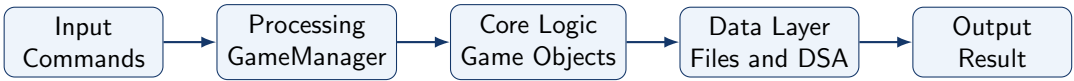
Problem Statement

Students practice OOP and Data Structures in separate small programs so they rarely see how the concepts work together. We are building a C++ detective game on our campus in which each concept has a real job.

Finalized Objectives

1. *Build a playable C++ detective game set on our campus in stages*
2. *Use OOP for the game objects and one data structure for each core feature*
3. *Give every member one module with individual GitHub commits*
4. *Verify each module with unit tests and a full playthrough*

Updated System Architecture / Workflow



Major Components

- *GameManager: runs the game*
- *Game objects: Map, Clue, Player*
- *Graph, stack, hash table, sorting*

Changes Since Phase-I

- *Console model first, SFML later*
- *Game logic separated from drawing*
- *Story loaded from text files*

Module Design & Functional Requirements

Major Modules

1. *Game flow (GameManager, Case)*
2. *World map (Location, Map)*
3. *Clues and suspects*
4. *Player and history*

Functional Requirements

- *Move between places; locked ones stay closed*
- *Search a place for clues; go back*
- *Look up a clue; sorted journal*
- *Accuse a suspect; evidence is checked*

Module Dependency

GameManager uses all modules. Player uses Clue and Stack. Map uses Location.

Technology Stack & Development Methodology

Technology Stack

- Programming: *C++*
- Database: *None (text files)*
- Framework: *SFML (Phase III)*
- Development Tool: *VS Code, g++*
- Version Control: *Git / GitHub*

Development Methodology

1. Requirement Analysis
2. System Design
3. Module Development
4. Integration
5. Testing & Validation

C++ supports both OOP and our own data structures. The story is kept in text files so no database is needed. We build in stages, one working version per phase.

Implementation of PBL Course Concepts

Course	Concepts Actually Implemented	Project Module
Data Structures	*Linked list (our Stack), Graph (adjacency list), Hashing with chaining, Sorting (insertion sort), Arrays (vector)*	*Map,Player,ClueTable, Journal*
OOPs with C++	*Classes, Encapsulation, Inheritance, Polymorphism, Abstraction, Templates, STL (vector, string)*	*All game objects*

Note: Only concepts that are actually implemented in our Phase-II model are listed.

Development & Implementation Progress

Module	Status	Work Completed
Game flow	Completed	*Story loading, command loop, accusation, hash table lookup*
World map	Completed	*Graph of 7 places, locked Exam Cell, go command*
Clues and suspects	Completed	*Clue family with polymorphism, insertion sort journal*
Player and history	Completed	*Player, linked-list Stack, back command*
Advanced features	Pending	*Shortest route, queue, keyword search, scoring*
2D interface	Pending	*SFML screens on top of the same game logic*

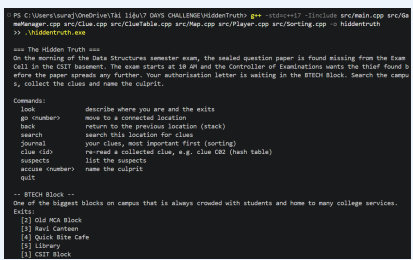
Overall Project Completion: *about 60%*

Working Prototype / Demonstration

Implemented Features

- *Walk between campus places*
- *Search places and collect clues*
- *Go back, look up clues, sorted journal*
- *Accuse a suspect and get a verdict*

Prototype / Output



Testing, Results & Validation

Test / Module	Expected Result	Observed Result	Status
Stack	*Last in, first out; empty pop is an error*	*Matches*	Pass
Hash table	*Every clue found, even in one bucket*	*Matches*	Pass
Graph	*Two-way routes; locked place opens*	*Matches*	Pass
Sorting, Player	*Sorted journal; back in reverse order*	*Matches*	Pass
Full playthrough	*Right culprit solves the case*	*Matches*	Pass

Results / Validation

All unit tests pass. The full case is solvable and a weak accusation is refused.

GitHub Repository & Development Activity

Repository Information

- GitHub: *Truth-Seekers09/the_detective_mystery*
- README.md: *Updated*
- Contributors: *All 4 members added*
- Repository Status: *Updated*
- Branches: *main (one branch per member from Phase 3)*

Development Activity

- *Small commits with clear messages*
- *Meaningful commit messages*
- *Individual contribution visible*
- *Latest implementation uploaded*
- *Documentation maintained*

Individual Contribution

Member	Role	Technical Contribution	GitHub Commits
Karan Badhani	*Team Lead*	*GameManager (loading, commands, accusation), Case, ClueTable hash table*	*30*
Khushi Singal	*Developer, Map*	*Location and Map: graph of 7 places, locked Exam Cell, go command*	*14*
SurajGiri Goswami	*Developer, Player*	*Player class and our own Stack (linked list): moveTo, goBack, journal*	*22*
Tanmay Arora	*Developer, Clues*	*Clue family (3 kinds), Suspect, insertion sort, sorting test*	*14*

Challenges, Remaining Work & Phase-III Roadmap

Challenges Faced

- *Pointers in our own stack and hash table*
- *Joining four dependent modules*
- *Making the story fair*

Solutions / Learning

- *Unit tests and turns on the shared test file*
- *How collisions and polymorphism work*

Phase-III Roadmap

1. *Shortest route (BFS), our own queue*
2. *Questioning suspects, keyword search*
3. *Scoring, more suspects and clues*
4. *SFML screens on the same logic*
5. *Final testing and documentation*

Expected Outcomes & References

Expected Final Outcomes

- *Complete working detective game*
- *Measurable results: unit tests and full playthrough*
- *A clear teaching example of OOP and DSA together*
- *Easy to extend: a new case is new data files*
- *Reusable code for future projects*

References

1. *C++ Reference (cppreference.com)*
2. *SFML Official Documentation*
3. *GeeksforGeeks: Data Structures*
4. *Microsoft Learn: C++ Documentation*
5. *Project repository on GitHub*

Thank You
Questions & Suggestions